

261. Which of the following property is not expected to be shown by copper

- (a) High thermal conductivity
- (b) Low electrical conductivity
- (c) Ductility
- (d) Malleability

Answer: B — (b) Low electrical conductivity

Chemical reason: Copper is an excellent electrical conductor, so “low electrical conductivity” is the property not expected.

Copper is also ductile, malleable, and a good thermal conductor.

262. Which of the following metal gives more than one chloride

- (a) Cu
- (b) Al
- (c) Ag
- (d) Na

Answer: A — (a) Cu

Chemical reason: This is a standard nomenclature/composition identification: the name or formula given in the question corresponds to Cu. The choice is fixed by the known composition of that compound/mineral.

263. The metal which is the best conductor of electricity is

- (a) Iron
- (b) Copper
- (c) Silver
- (d) Aluminium

Answer: C — (c) Silver

Chemical reason: Silver has the highest electrical conductivity of all metals at ordinary temperature, slightly higher than copper; therefore Ag is the best conductor among the options.

264. Paris green is

- (a) Double salt of copper carbonate and copper nitrate
- (b) Double salt of copper acetate and copper arsinite
- (c) Double salt of copper acetate and copper sulphate
- (d) Double salt of copper and silver nitrate

Answer: B — (b) Double salt of copper acetate and copper arsinite

Chemical reason: Paris green is copper acetoarsenite, historically represented as a double salt/compound derived from copper acetate and copper arsenite. It is a bright green but highly toxic pigment.



265. Reaction between the following pairs will produce H_2 except

- (a) Na^+ ethyl alcohol
- (b) Fe^+ steam
- (c) $Fe + H_2SO_4$ (aq.)
- (d) $Cu + HCl$ (aq.)

Answer: D — (d) $Cu + HCl$ (aq.)

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives Cu^+HCl (aq.). The choice is consistent with the expected oxidation state and stability of the product.

266. Which of the following is correct

- (a) Gun metal : $Cu + Zn + Sn$
- (b) Duralumin : $Al + Cu + Mg + Ag$
- (c) German silver : $Cu + Zn + C$
- (d) Solder : $Pb + Al$

Answer: A — (a) Gun metal : $Cu + Zn + Sn$

Chemical reason: The correct choice is Gun metal : $Cu + Zn + Sn$. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

267. Solder is an alloy of

- (a) 70% lead, 30% tin
- (b) 30% lead, 70% tin
- (c) 80% lead, 20% tin
- (d) 90% Cu, 10% tin

Answer: B — (b) 30% lead, 70% tin

Chemical reason: This is determined by the standard composition or heat-treatment

behaviour of the metal/alloy. The description in 30% lead, 70% tin matches the relevant metallurgical process or alloy property.

268. Zinc forms two important alloys, (i) Brass and (ii) German silver. Metals present in them mainly are

- (a) In (i) zinc and tin; and in (ii) zinc, silver and nickel
- (b) In (i) zinc and iron; and in (ii) zinc, nickel and cobalt
- (c) In (i) zinc and copper; and in (ii) zinc, copper and nickel
- (d) In (i) zinc and aluminium; and in (ii) zinc, nickel and aluminium

Answer: C — (c) In (i) zinc and copper; and in (ii) zinc, copper and nickel

Chemical reason: Brass is $Cu-Zn$, while German silver is $Cu-Zn-Ni$. Thus the common correct compositions are zinc + copper for brass and zinc + copper + nickel for German silver.

269. One of the constituent of german silver is

- (a) Ag
- (b) Cu
- (c) Mg
- (d) Al

Answer: B — (b) Cu

Chemical reason: Silver chemistry is governed strongly by low-solubility halides and complex formation with ligands such as NH_3 or $S_2O_3^{2-}$. Under the stated conditions this leads to Cu.



270. Gun metal is an alloy of

- (a) Cu and Al
- (b) Cu, Sn and Zn
- (c) Cu, Zn and Ni
- (d) Cu and Sn

Answer: B — (b) Cu, Sn and Zn

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Cu, Sn and Zn matches the relevant metallurgical process or alloy property.

271. Besides Zn and Cu, german silver contains the metal

- (a) Sn
- (b) Ag
- (c) Ni
- (d) Mg

Answer: C — (c) Ni

Chemical reason: Silver chemistry is governed strongly by low-solubility halides and complex formation with ligands such as NH_3 or $\text{S}_2\text{O}_3^{2-}$. Under the stated conditions this leads to Ni.

272. Which metal is present in brass, bronze and German silver

- (a) Zn
- (b) Mg
- (c) Cu
- (d) Al

Answer: C — (c) Cu

Chemical reason: Copper is present in brass, bronze, and German silver. Brass is Cu–Zn, bronze is Cu–Sn, and German silver is Cu–Zn–Ni.

273. Which of the following is wrongly matched

- (a) German silver Cu+Zn+Ni
- (b) Alnico Fe+Al+Ni+CO
- (c) Monel metal Cu+Zn+Sn
- (d) Duralumin Al+Cu+Mg+Mn

Answer: C — (c) Monel metal Cu+Zn+Sn

Chemical reason: The correct choice is Monel metal $\text{Cu}^+\text{Zn}+\text{Sn}$. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

274. An extremely hot copper wire reacts with steam to give

- (a) CuO
- (b) Cu_2O
- (c) Cu_2O_2
- (d) CuO_2

Answer: B — (b) Cu_2O

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives Cu_2O . The choice is consistent with the expected oxidation state and stability of the product.



275. From a solution of CuSO_4 , the metal used to recover copper is

- (a) Sodium
- (b) Iron
- (c) Silver
- (d) Hg

Answer: B — (b) Iron

Chemical reason: Iron is more reactive than copper and displaces Cu^{2+} from copper salt solutions: $\text{Fe} + \text{Cu}^{2+} \rightarrow \text{Fe}^{2+} + \text{Cu}$. Therefore iron is used to recover copper from solution.

276. Copper sulphate is commercially made from copper scraps by

- (a) Dissolving in hot conc. H_2SO_4
- (b) The action of dil. H_2SO_4 and air
- (c) Heating with sodium sulphate
- (d) Heating with sulphur

Answer: B — (b) The action of dil. H_2SO_4 and air

Chemical reason: Copper sulfate can be prepared industrially from copper scrap by oxidation in dilute sulfuric acid in the presence of air/oxygen, which supplies the oxidizing power needed because Cu does not liberate H_2 from dilute acid.

277. Cuprous ion is colourless, while cupric ion is coloured because

- (a) Both have unpaired electrons in d -orbital
- (b) Cuprous ion has a completed d -orbital and cupric ion has an incomplete d -orbital
- (c) Both have half-filled p and d -orbitals
- (d) Cuprous ion has incomplete d -orbital and cupric ion has a completed d -orbital

Answer: B — (b) Cuprous ion has a completed d -orbital and cupric ion has an incomplete d -orbital

Chemical reason: Cu^+ has a $3d^{10}$ configuration, so there is no partially filled d subshell for d-d transitions and its compounds are often colourless. Cu^{2+} is $3d^9$ and usually coloured.

278. A blue colouration is not obtained when

- (a) Ammonium hydroxide dissolves in copper sulphate
- (b) Copper sulphate solution reacts with $\text{K}_4[\text{Fe}(\text{CN})_6]$
- (c) Ferric chloride reacts with sodium ferrocyanide
- (d) Anhydrous CuSO_4 is dissolved in water

Answer: B — (b) Copper sulphate solution reacts with $\text{K}_4[\text{Fe}(\text{CN})_6]$

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in Copper sulphate solution reacts with $\text{K}_4[\text{Fe}(\text{CN})_6]$ has the characteristic oxidation state/ligand field that produces the observed colour.





279. Identify the statement which is not correct regarding CuSO_4

- (a) It reacts with KI to give iodine
- (b) It reacts with KCl to give Cu_2Cl_2
- (c) It reacts with NaOH and glucose to give Cu_2O
- (d) It gives CuO on strong heating in air

Answer: B — (b) It reacts with KCl to give Cu_2Cl_2

Chemical reason: The correct choice is It reacts with KCl to give Cu_2Cl_2 . It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

280. Copper sulphate solution reacts with KCN to give

- (a) $\text{Cu}(\text{CN})_2$
- (b) CuCN
- (c) $\text{K}_2[\text{Cu}(\text{CN})_4]$
- (d) $\text{K}_3[\text{Cu}(\text{CN})_4]$

Answer: D — (d) $\text{K}_3[\text{Cu}(\text{CN})_4]$

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives $\text{K}_3[\text{Cu}(\text{CN})_4]$. The choice is consistent with the expected oxidation state and stability of the product.

